The Population Estimates Program

The number you get when nobody counted, and the arithmetic that produces it

84-355 Democracy's Data

August 2026

Almost every sentence that begins "the population of X is" is quoting neither of the two instruments in the last two chapters. The decennial census is ten years old the day after it is taken. The American Community Survey is a sample. Neither can tell you how many people live in a county *this year*.

The Population Estimates Program can, and it does it without asking anybody anything. It takes the last census and carries it forward with arithmetic. That makes it the third instrument, and the one people quote most while knowing least about. As the last chapter found, that includes anyone looking up an ACS county population, which is forced to agree with this program's number.

01 The whole method is one line of arithmetic

This year's population is last year's, plus the people born, minus the people who died, plus the people who moved in, minus the people who moved out.

That is not a simplification of the method. That *is* the method, and every term in it is published. Which means the claim is checkable — so here it is, checked on every county in the country:

Consistency check	Counties checked	Violations
Last year + natural change + net migration + residual = this year	3,144	0
International + domestic migration = net migration	3,144	0
Births - deaths = natural change	3,144	0

Zero violations in 3,144 counties. Every published estimate is exactly the sum of the published components beneath it.

That is worth doing rather than believing, because of what it establishes about the object. A census figure is the result of an attempt to find people. An ACS figure is the result of asking some of them. **A population estimate is the result of an addition** — and if any input to that addition is wrong, the total is wrong in a way no amount of staring at the total will reveal.

02 What arrives

```
One row of the file: DeKalb County, Georgia.
The file is 1.8 MB of plain CSV and every county carries the same
columns. These are the ten that make up one year's arithmetic:

  POPESTIMATE2023            768,047
  BIRTHS2024                  10,016
  DEATHS2024                   5,200
  NATURALCHG2024               4,816
  INTERNATIONALMIG2024         8,755
  DOMESTICMIG2024            -11,219
  NETMIG2024                  -2,464
  RESIDUAL2024                   -92
  NPOPCHG2024                  2,260
  POPESTIMATE2024            770,307

Read down: last year's estimate, then births and deaths and the
difference between them, then the two halves of migration and their
sum, then the residual, then the total change, then this year's
estimate. The last line is the sum of everything above it.

STATE and COUNTY are zero-padded FIPS codes and must be read as
text. As numbers they lose their leading zeros and stop joining to
anything else in this book.
```

> **Before you read on, commit to a number.** In 2024 the population of the United States grew by about 3.3 million people. **What share of that growth do you think came from births exceeding deaths, and what share from migration?**

03 What the components said in 2024

Component	People
Births	3,605,563
Deaths	-3,086,925
Natural change	518,638
International migration	2,786,119
Domestic migration	0
Net migration	2,786,119
Residual	0
Total change in 2024	3,304,757

Two rows deserve a second look.

Domestic migration is exactly zero. Not approximately — exactly. That is not a finding about America, it is arithmetic: every person who leaves one county arrives in another, so internal movement nets to nothing nationally. Domestic migration redistributes the population and adds none of it. It is the single largest force reshaping individual counties and it is invisible at the national level.

International migration is 2,786,119 of 3,304,757. That is **84.3% of all United States population growth in 2024**, against 518,638 from births exceeding deaths. The country grew mostly because people arrived, not because more were born than died.

04 Two-thirds of counties had more deaths than births

Quantity	Value
Counties with more deaths than births, 2024	2,076
Share of all counties	66.0%
People living in those counties	97,700,142
Share of the country living in them	28.7%

2,076 counties recorded more deaths than births in 2024, or 66.0% of them. Natural decrease is not a rare condition of a few dying places. It is the ordinary state of two-thirds of American counties.

Those counties hold only 28.7% of the population, That is the other half of the same fact. The counties where deaths outnumber births are mostly small ones, and the country's people are concentrated in the minority that still has more births.

Quantity	Counties
Counties that grew in 2024	2,054
Of those, would have shrunk without international migration	451
Of those, would have shrunk without natural change	91
Counties that grew despite more deaths than births	1,160

Read those together. 2,054 counties grew. **1,160 of them grew while burying more people than they delivered** — migration made up the difference. And 451 of the counties that grew **would have shrunk without international migration specifically**.

That last figure is subtraction, not modeling: a county's published total change, minus its published international migration. Nothing is re-estimated. It says what the file says.

05 The part that does not add up

The identity above has a term this chapter has not yet explained. Alongside births, deaths and migration, the Bureau publishes a column called **residual** — what is left over, the part of the change the components do not account for.

Quantity	Value
Counties with a residual of zero	663
Counties with a non-zero residual	2,481
Largest residual, in people	247
Total residual, all counties	0
Total residual as a share of total change	0.0%

2,481 of 3,144 counties carry one, and the largest is 247 people. But look at the total: **the residual sums to 0 across the whole country.** It is constrained to net out. It moves people between counties to make each county's arithmetic close, and adds none to the national total.

That should sound familiar. It is the same shape as the disclosure-avoidance noise in the source chapter: an adjustment that is real at the bottom of the hierarchy and vanishes at the top, held there by construction. The difference is that this one is printed in its own column, and you can read it.

A residual is not a scandal. It is what an accounting identity does when the components come from different administrative systems that do not perfectly agree. It is published rather than absorbed silently into migration, which is the honest choice. But it is a plug, and it is worth knowing that the tidiest of the three instruments has one.

06 Why this chapter matters to the other two

The estimates are not an alternative that careful people avoid. They are *inside* the data most people use.

When you look up a county's population in the ACS, the number you get was controlled to this program. The survey was not allowed to disagree with it. That is why 3,090 of 3,222 ACS county rows print no margin of error, no statement of how far off the estimate could be, as the ACS chapter found. The estimate is doing work in places that do not announce it.

So the three instruments are not three independent readings you can average or choose between. The decennial anchors the estimates; the estimates control the survey. A problem in the census propagates forward into both.

07 What this chapter cannot tell you

One vintage, one year. Everything here is the 2024 vintage, describing change in 2024. The Bureau reissues the whole series every year, and a reissued vintage revises earlier years — so a "2023 population" from this file and a "2023 population" from last year's file are different numbers. This chapter does not measure that drift; it is the most important thing about estimates that it leaves out.

The components are not measured here, only used. Births and deaths come to the Bureau from vital-statistics systems, and migration from a mix of administrative records and survey data. This chapter checks that the published components sum to the published totals. It cannot check whether any component is right, and the identity holding tells you nothing about that — a sum of wrong numbers adds up perfectly well.

Counties are as far down as this goes. The program publishes estimates for places below the county level too, and the smaller the place the more the estimate depends on assumptions rather than on records.

"Would have shrunk without international migration" is not a forecast. It subtracts one published component from one published total for one year. It is not a claim about what people would have done had they not moved, or about any future year.

08 Sources

- U.S. Census Bureau. *Population Estimates Program*, county totals with components of change, vintage 2024 (`co-est2024-alldata.csv`). https://www2.census.gov/programs-surveys/popest/datasets/2020-2024/counties/totals/
- The −555555555 sentinel that marks a controlled estimate is in the ACS chapter. The control relationship itself is measured in the access chapter, which finds a county's one-year survey total to be this file's estimate exactly.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`components.csv`, `growth.csv`, `identity.csv`, `natural.csv`, `residual.csv`, `controlled.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `identity.csv` records that the accounting identity holds in every county, with no violations at all. A check that cannot fail teaches nothing. Work out what would have to go wrong in the world for that row to break.
- `residual.csv` counts the counties whose residual is exactly zero. The residual is the part the arithmetic could not explain. Does a zero mean nothing was unexplained, or that nothing was measured?
- `natural.csv` says most counties buried more people than they delivered. `growth.csv` counts the counties that grew anyway. Hold the two side by side and explain how both can be true.

- `components.csv` gives the national totals going into the estimate. Which component is largest, and which one can the Bureau least afford to get wrong?

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

```
I want to rebuild a public dataset from its original source, without any
starting files. Please do the following and show your work.
THE SOURCE. U.S. Census Bureau, Population Estimates Program, county
totals with components of change, vintage 2024. One CSV, about 1.8 MB,
published at
https://www2.census.gov/programs-surveys/popest/datasets/2020-
2024/counties/totals/co-est2024-alldata.csv
No account or key is needed. Each row is a state or county; the SUMLEV
column says which (040 is a state, 050 is a county). Read the STATE and
COUNTY columns as text, not numbers — they are zero-padded codes, and
"01" stops being Alabama the moment it becomes 1.
WHAT TO BUILD.
1. Check the Bureau's own accounting identity on every county: the 2023
estimate, plus births minus deaths, plus net migration, plus the
column called RESIDUAL, should equal the 2024 estimate exactly.
Report how many counties pass and how many fail.
2. Count the counties where deaths exceeded births in 2024, and what
share of all counties that is.
CHECK YOUR WORK. The copy this book used was fetched in August 2026.
If your rebuild matches it, you should find:
- 3,195 rows in the file: 51 state rows and 3,144 county rows
- the identity holds on all 3,144 counties, with zero violations
- 2,076 counties had more deaths than births in 2024 — 66% of them
The Bureau replaces this file when a new vintage comes out each spring;
if the URL now says a later year, the numbers will differ because the
estimates were revised, not because you made an error.
```